MODULE *BaselineMutex*

EXTENDS *Naturals*, *FiniteSets*

CONSTANT *Proc*

ASSUME $Proc \neq \{\}$

VARIABLES *pc*, *mutex*

$Vars \triangleq \langle pc, mutex \rangle$

PCs:
"ncs" : non-critical section
"trying" : attempting to acquire *mutex*
"cs" : inside critical section
"exit" : leaving critical section and releasing *mutex*

$$
\begin{aligned}
&Init \triangleq \\
&\quad \land pc = [i \in Proc \mapsto \text{"ncs"}] \\
&\quad \land mutex = 1
\end{aligned}
$$

$$
\begin{aligned}
&EnterTrying(i) \triangleq \\
&\quad \land pc[i] = \text{"ncs"} \\
&\quad \land pc' = [pc \text{ EXCEPT } ![i] = \text{"trying"}] \\
&\quad \land mutex' = mutex
\end{aligned}
$$

$$
\begin{aligned}
&AcquireMutex(i) \triangleq \\
&\quad \land pc[i] = \text{"trying"} \\
&\quad \land mutex = 1 \\
&\quad \land pc' = [pc \text{ EXCEPT } ![i] = \text{"cs"}] \\
&\quad \land mutex' = 0
\end{aligned}
$$

$$
\begin{aligned}
&LeaveCS(i) \triangleq \\
&\quad \land \quad pc[i] = \text{"cs"} \\
&\quad \land \quad pc' = [pc \text{ EXCEPT } ![i] = \text{"exit"}] \\
&\quad \land \quad mutex' = mutex
\end{aligned}
$$

$$
\begin{aligned}
&ReleaseMutex(i) \triangleq \\
&\quad \land pc[i] = \text{"exit"} \\
&\quad \land mutex = 0 \\
&\quad \land pc' = [pc \text{ EXCEPT } ![i] = \text{"ncs"}] \\
&\quad \land mutex' = 1
\end{aligned}
$$

$$
\begin{aligned}
&ProcStep(i) \triangleq \\
&\quad EnterTrying(i) \\
&\quad \lor AcquireMutex(i) \\
&\quad \lor LeaveCS(i) \\
&\quad \lor ReleaseMutex(i)
\end{aligned}
$$

$Next \triangleq$
$\quad \exists\, i \in Proc : ProcStep(i)$

$TypeOK \triangleq$
$\quad \land \quad pc \in [Proc \rightarrow \{\text{"ncs"}, \text{"trying"}, \text{"cs"}, \text{"exit"}\}]$
$\quad \land \quad mutex \in \{0, 1\}$

$InCS \triangleq$
$\quad \{i \in Proc : pc[i] = \text{"cs"}\}$

$MutualExclusion \triangleq$
$\quad Cardinality(InCS) \leq 1$

$Spec \triangleq$
$\quad Init \land \Box[Next]_{Vars}$